MIL-W-80117B
22 November 1985
SUPERSEDING
MIL-W-80117A
6 March 1978

MILITARY SPECIFICATION

WELDING SETS, SUBMERGED ARC

This specification is approved for use by all Departmental Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers submerged arc welding sets using flux shielding and consumable electrodes in conjunction with equipment and accessories specified herein.

1.2 Classification. The welding sets shall be of the following types, sizes, and classes. The types, sizes, and classes to be furnished shall be as specified (see 6.2.1).

Type I - Automatic
Type II - Semi-automatic
Type III - Multi-wire (automatic)

Size 1 - 500 ampere capacity
Size 2 - 600 ampere capacity
Size 3 - 800 ampere capacity
Size 4 - 900 ampere capacity
Size 5 - 1200 ampere capacity
Size 6 - 1500 ampere capacity
Size 7 - 2000 ampere capacity

Class A - Stationary mounting
Class B - Beam carriage mounting
Class C - Wheel carriage (self propelled) mounting
Class D - Wheel carriage mounting
Class E - Hand portable mounting

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, Memphis, Tennessee 38114-5297, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3431
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:39Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-W-45562 - Welding and Soldering Equipment, Supplies and Accessories, Packaging of.

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards for Federal Services.

FED-STD-376 - Preferred Metric Units for General Use by the Federal Government.

2.1.2 Other Government documents and publications. The following other Government document forms a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of solicitation.

U. S. DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910).

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which are current on the date of the solicitation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/NEMA EW 1 - Electric Arc-Welding Apparatus.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order (see 6.2.1), a sample shall be subjected to first article inspection (see 4.4 and 6.3).

3.2 Design. The welding sets shall be new and one of the manufacturer's current models capable of submerged arc welding and rough usage in accordance with the requirements herein. The welding sets shall include all components, parts, and features necessary to meet the performance requirements specified herein. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement, and repair. Type I and type III units shall have all components capable of being compactly secured on the mounting device and arranged for automatic operation, and type II units shall have all components except the gun and cable/hose assembly compactly secured on the mounting device and arranged for semi-automatic operation. The welding sets shall conform to applicable requirements of NEMA EW-1.

3.2.1 Materials. Materials shall be as specified herein. Materials not specified herein shall be selected by the manufacturer to meet the requirements of this specification. Contact between dissimilar metals shall be avoided whenever possible. Materials shall be free from all defects that affect the serviceability and appearance of the finished product.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

3.2.1.1 Reclaimed materials. The welding sets may contain reclaimed materials. The reclaimed materials shall have been reprocessed, remanufactured, or recycled in a manner which will restore the material to the same chemical composition and physical properties as the materials originally selected for use on the welding sets.

3.2.2 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances which will permit installation of replacement parts without modification of parts or machine.

3.2.3 Measurement systems. Unless otherwise specified, either the U.S. Customary System of Units (US) or the International System of Units (SI) shall be used in the design and construction of the welding set. When only one system of measurements is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes, and capacities are given in SI units (with US system in parenthesis). These measurements may be verified through the use of the conversion factors and methods specified in FED-STD-376.

3.2.4 Energy efficiency. The welding set and its applicable components that directly consume energy in normal operation shall be designed and constructed for the highest degree of energy efficiency as governed by the latest developments available within the industry.

3.2.5 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the welding set that present safety hazards. The safety devices shall not interfere with the operation of the welding sets. The safety devices shall prevent unintentional contact with the guarded part, and shall be removable to facilitate inspection, maintenance, and repair of the parts. All welding set parts, components, mechanisms, and assemblies furnished on the welding sets, whether or not specifically required herein, shall comply with all of the requirements of OSHA 2206 that are applicable to the welding set itself. Additional safety and health requirements shall be as specified (see 6.2.1 and 6.4).

3.2.6 Mercury restriction. The welding set shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.7 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the welding sets.

3.2.8 Environmental protection. The welding set shall be so designed and constructed that, under the operating, service, transportation, and storage conditions described herein, the welding set shall not emit materials hazardous to the ecological system as prescribed by federal, state, or local statutes in effect at point of installation (see 6.4).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

3.2.9 Threads. All threaded parts used on the welding sets and related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3 Construction. The welding set shall be constructed of parts which are new, without defects, and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the welding set at its maximum rating and capacity without permanent distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defect. Processes such as welding, peening, plugging, or filling with solders or pastes shall not be used on castings or forgings to reclaim any defective components for use on the welding sets. Such processes may be used only for enhancing surface finish and appearance.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. None of these processes shall be employed as a repair measure for any defective part.

3.3.3 Fastening devices. All screws, pins, bolts, and other fasteners shall be installed in a manner to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be swaged, peened, staked, or otherwise permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of sand, dirt, fins, sprues, flash, scale, flux, and other harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

3.3.5 Painting. Unless otherwise specified (see 6.2.1), the welding sets shall be painted in accordance with the manufacturer's commercial practice, provided that all surfaces to be painted shall be cleaned of all foreign matter and at least one coat of primer and one coat of finish color are applied.

3.3.6 Accessibility. The welding sets shall be provided with removable panels or doors for access to terminal boards and other parts which may require replacement or adjustment. Access panels for current carrying components shall be removable only with the use of tools. Components within the cabinet shall be arranged so that they may be removed or replaced without undue interference from, or removal of, or damage to, other components and wirings.

3.3.7 Compatibility. Due to the different configuration of the various components, accessories, and optional features, the supplier shall furnish

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

and, as necessary, install all such electrical and mechanical devices necessary to assure that all components, accessories, and optional features required shall be compatible as required to assure the desired performance.

3.3.8 Limiting devices. When specified (see 6.2.1), limiting devices to stop or to stop and reverse the movement of self-propelled mounting devices (see 3.5.5) shall be furnished. When such devices are included, the required effect of these devices on the control requirements for the welding sets shall be included (see 3.5.2).

3.4 Performance. When connected to the proper power source and operated in accordance with the manufacturer's printed instructions for arc welding applications, the welding sets shall produce satisfactory welds within the capacities and requirements specified herein. The welding sets shall be capable of operating continuously at rated output without exceeding the designed operating conditions of any of the components. The welding sets shall be capable of utilizing and controlling the range of input currents of the types as shown to effectively perform submerged arc welding with the sizes of wires at the range of wire speeds as shown in table I.

3.5 Components. The welding sets covered by this specification shall consist of, but not be limited to, a power supply unit, a control unit, a wire feeder, a flux hopper, a mounting device, enclosures, interconnecting cables and hoses. A gun and cable-hose assembly for type II shall be included.

3.5.1 Power supply unit. The power supply shall conform to the requirements of an existing military specification as specified (see 6.2.1). The power supply shall furnish the required welding power to meet applicable requirements of the wire feeder unit and control unit.

3.5.2 Control unit. The control unit for the welding sets shall provide the necessary degree of control to assure that all components of the welding sets operate properly and in the desired sequence. Controls as used with the power carriage shall provide for constant speed of travel. Operating controls and switches and indicator lights and meters shall be centrally located convenient to the operator. All controls necessary to assure proper starting operation and manipulation of the components and all meters necessary to indicate operating conditions shall be furnished and installed. The type II (semi-automatic) sets shall include, as a minimum, a wire feed speed control and a wire feed inching control; and the type I and type III (automatic) sets shall include, as a minimum, a contactor or field relay with start-stop push button or switch, a power on-off switch and indicator light, a flux flow control switch, wire feed speed control, wire inching control, voltage and current control, and a voltmeter and an ammeter.

3.5.3 Wire feeder unit. The wire feeder unit shall properly feed the welding wire to the weld zone at controlled rates. The feeder shall consist basically of a wire feed motor, gear reduction unit, wire drive, contact devices, accessory support and mounting clamp, and accessories. The wire feeder unit shall be compatible with the control unit, with the welding power supply, and with the mounting device. The unit shall have flexibility

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

TABLE I - Submerged arc welding sets.

Size	Type	Welding Current		Range of Wire Sizes cm (Inches)	Wire Speeds mm (IPM)	Mounting (Class)	Speed of Mounting CM/M (IPM)		Hopper Capacity kg (Pounds)
		Range	Ampere				Class B	Class C	
1	II	100-500	AC-DC	.198-.238 (5/64-3/32)	1.52-6.6 (60-260)	E			1.8 (3)
2	II	120-600	DC	.08-.238 (.035-3/32)	.81-12.5 (50-500)	C-D-E			45.4 (100)
3	II	250-800	AC-DC	19.85-.238 (5/64-3/32)	1.9-5.74 (75-226)	D			1.8 (4)
4	I	250-900	AC-DC	.238-.48 (3/32-3/16)	1.9-5.74 (75-226)	A-B-C	7.6-203 (5-75)	12.8-127 (7-50)	9 (20)
5	I	250-1200	AC-DC-DC-CV	.16-.64 (1/16-7/32)	1.27-4.28 (50-169)	A-B-C	7.6-203 (5-75)	12.8-127 (7-50)	9 (20)
6	I	300-1500	AC-DC	.238-.56 (3/32-7/32)	.5-5.08 (20-200)	A-B-C	15.2-132 (6-52)	15.2-234.8 (6-96)	9 (20)
7	I	450-2000	AC-DC DC-CV	.16-.8 (1/16-5/16)	2.69-5.55 (106-219)	A-B-C	7.6-203 (5-75)	38.1-266.7 (15-105)	9 (20)

NOTE: Type III welding sets shall consist of combinations of sizes 4 thru 7 as required (see 3.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

allowing the unit or its necessary components to be raised or lowered, moved laterally, or swiveled to meet various welding applications. The wire feeder unit shall stop the feed of the wire when the wire feed switch is put in the "stop" position.

3.5.3.1 Wire drive. The wire drive unit shall be driven by a gear reduction unit and shall consist of all necessary devices to properly move the wire. The drive shall consist of one or more feed rolls and pressure rolls. The feed rolls shall be of the friction type. The wire drive unit shall be capable of feeding all sizes of wire specified in table I by manual shifting of gears or by changing sizes of either the gears or rolls. The pressure rolls shall be utilized to increase or decrease the pressure of the feed rolls on the wire, and adjustment of this pressure shall be a simple operation. The feed rolls and pressure rolls shall be electrically insulated from the exterior of the wire drive unit.

3.5.3.2 Contact devices. The contact devices shall be used to conduct the welding power into the welding wire and shall consist of the busbar assembly, the contact nozzle assembly, and other components. The busbar assembly shall be separated from the wire feeder unit housing by adequate insulation blocks. The contact nozzle assembly shall transmit the welding power from the busbar into the welding wire and shall consist of wire guides sized from the wire in use, the nozzle, and the insulating nozzle. The wire guides shall be easily replaced. When specified (see 6.2.1), a complete set of guides shall be furnished for each size required.

3.5.3.3 Accessory support and mounting clamp. The housing or base of the wire feeder unit shall include the necessary means for accessory support and mounting clamp to allow the unit to support, carry, or be attached to the other components and accessories such as the mounting device, the control unit, and the wire reel.

3.5.3.4 Wire holder. The sets shall have provisions for mounting a wire reel. Installation of wire reel shall be accomplished without the use of tools. The holder shall keep the coil concentric with the hub and provide a smooth pay-off of wire, and shall have an adjustable device to put drag or tension on the hub so that the reel will stop turning when the wire feed stops. The wire holder shall be electrically insulated from the frame of the unit. The arms of the holder shall be adaptable to accommodate a variety of wire coils of different inside diameter and widths. The holder shall be of such construction to provide a firm support for the various sizes and types of wire coils.

3.5.3.5 Wire straightener. The welding sets shall be furnished with a wire straightening device. The straightener shall remove kinks and bends from the welding wire. Components shall be easily replaceable.

3.5.4 Flux hopper. The flux hopper shall be abrasion resistant material and construction. Design of the hopper shall be such to allow easy loading and a smooth flow of flux into the flux hose or into the gun. Regulation of

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:39Z
Check the source to verify that this is the current version before use.

the flow of flux shall be provided. The storage capacity of the hopper shall be not less than the amount shown in table I.

3.5.4.1 Flux valve. The manual or solenoid operated automatic flux valve shall provide a simple and convenient control of the flow of flux to the welding zone. Design and location shall be such to assure stopping of the flow of flux when the valve is closed. The discharge of flux from the flux valve on type I sets shall pass through a spout. The spout shall be such to allow a further regulation of the shape and depth of the flux deposit. The type I and type III welding sets shall have an automatic flux valve (except class C) with controls tied-in with the control unit, and the type II welding sets shall have manually operated flux valves to start and stop the flux flow.

3.5.4.2 Pneumatic flux feed system. The size 2 set shall include, but not be limited to, a pneumatic flux feed system consisting of an air tank, air filter, pressure regulator, flux hose, and gun. The tank shall have a wide mouth to facilitate easy filling and shall have a capacity of 45.4 kg (100 pounds) of flux. Operation shall be such that when connected to a source of 4.2 - 8.4 kg/cm (60 to 120 psi) pressure, a smooth flow of flux flows from the hopper through the hose to gun shall occur when the gun nozzle is pointed down, and shall stop when the gun nozzle is tipped up. The compressed air shall act only to force the flux to the nozzle rather than to carry it, thus requiring a minimum consumption of air. The gun as furnished with this pneumatic system shall be compatible with the other functions of the welding set. The pneumatic flux feed system shall supply and distribute the proper quantities of flux to meet the requirements of the welding set. The flux feed hose shall be at least 4.6 m (15 feet) long.

3.5.5 Mounting devices. The welding sets shall have one of the following mounting devices (see 1.2).

3.5.5.1 Class A stationary mounting. The stationary mounting device shall be designed to rigidly hold the welding set and designed to allow submerged arc welding to moving components under the welding head.

3.5.5.2 Class B beam carriage. The beam carriage mounting device shall be designed for straight line welding. The carriage shall be driven by an electric motor and shall have controls to move forward, backward, start and stop. Speeds of travel and length of travel shall be as specified (see 6.2.1).

3.5.5.3 Class C wheeled carriage (self-propelled). The class C wheeled carriage assembly shall consist of a housing, driving motor, drive train, controls, and other features necessary to meet the requirements of this specification. The carriage housing shall be designed to completely protect or enclose the drive components. The carriage shall be mounted on two drive wheels and one or two caster wheels. Caster wheel shall be free to pivot about a vertical axis and shall be lockable for straight line motion. A mounting arm with provision for adjustment to accommodate the other com-

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

ponents shall be attached to the top of the carriage housing. The carriage shall be capable of carrying operating loads in excess of 65.8 kg (145 pounds). The drive train shall consist of driving wheels and transmission assembly connected through a clutch to the motor. The driving wheels shall be steel, and have a knurled periphery to aid positive traction, and shall be hardened for long wear. Controls shall be furnished to change direction and speed of carriage travel, and to disengage the drive train. The controls shall be mounted in the carriage housing in a position convenient to the operator. A switching means, indicating "Forward", "Off", and "Reverse" shall control direction of carriage travel. An indexed speed control shall provide variable adjustment of travel speed by dialing the control knob. Speed adjustment shall be possible while the carriage is in motion. A clutch lever shall engage or disengage the clutch in the drive train. As required, the carriage control devices shall be adjustable with and controllable from the welding set control unit.

3.5.5.4 Class D wheel carriage. The class D wheel carriage shall be of the non-powered type vehicle with three or four rubber tired wheels. The front wheel or wheels shall be free to swivel while the back wheels shall be fixed. The base of the carriage shall be of such design and construction to support and secure the components of the weld set in such a manner to have a low center of gravity. As required, a suitable handle type tow bar shall be furnished and installed for guiding and moving the carriage.

3.5.5.5 Class E portable unit. The class E portable mounting device shall be of the hand carry type of light construction including a suitable base and adequate handles. The components shall be suitably arranged and adequately secured to the base in such a manner to facilitate smooth operation and simple movement from one location to another.

3.5.6 Enclosure. The enclosure shall be constructed to support and protect contents of welding sets. Access shall be provided for servicing of enclosed parts. Ventilation shall be provided to assure proper operating temperatures.

3.5.7 Interconnecting cables and hoses. The welding sets shall have electric cords and cables, welding cables, and flux hoses from the proper interconnection of all components. All cords and cable shall be adequately fastened to withstand normal strains, and flexibly mounted to accommodate all movements of the various components. The flux hoses shall be of the abrasion resistant type, properly fastened to withstand normal strains, and flexibly mounted to accommodate the normal changing positions utilized. The sizes and lengths of all cables and hoses shall meet all requirements of this specification. When required to meet special conditions, the lengths of all cables and hoses shall be as specified (see 6.2.1).

3.5.8 Gun and cable assembly (type II only). The gun and cable assembly as part of the type II welding sets shall consist of a hand-held gun with a straight or pistol type grip, a welding source power cable, a flexible conduit for feeding and guiding the electrode wire, and a flux supply device

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

consisting of either a gun mounted flux hopper or a flux supply hose. The cable assembly shall be at least 4.5 m (15 feet) long and shall connect the components of the welding set to the gun. The gun shall be constructed of lightweight, high-impact-resistant material, and shall be designed to be air cooled, and to be manually operated with one hand. All of the outside surfaces shall be electrically insulated from the welding and control current. The gun shall be designed for the cables and hoses to be attached at the rear or back of the body. Flux hoppers shall have a capacity to hold 1.8 kg (4 pounds) of flux. The guns shall have a replaceable nozzle to direct the flow of shielding flux to completely shield the electrode and work from the surrounding atmosphere. The gun shall have a replaceable contact tip or curved tube to transfer the full amperage of the welding current to the electrode. The controls to operate the gun shall be incorporated in the gun or in the control unit. Means shall be provided to start and stop the wire feed, the flow of shielding flux, the welding current, and to inch the wire. The cable hose assembly shall consist of the welding current conductor, the shielding flux hose (as required), wire guide conduit assembly, and electric control cable. The cable assembly shall be enclosed in a heat and abrasion-resistant cover.

3.6 Optional equipment. When required (see 6.2.1), optional equipment (a magnetic separator, high frequency generator, oscillator assembly, or any other equipment) shall be furnished as specified.

3.6.1 Unfused flux recovery systems and fused flux reclaiming system. When required unfused flux recovery systems or fused flux reclaiming systems shall be provided as specified (see 6.2.1).

3.7 Multiple welding sets. Multiple welding sets (type III) shall consist of two or more welding heads to provide two or more consumable electrodes feeding into the weld zone(s). The multiple sets shall be combinations of the sizes of sets shown in table I arranged to provide the necessary physical configuration to control and supply the proper power, to control and feed the proper electrodes and flux, and to control and carry all of the components. The components of the multiple welding sets shall meet the applicable requirements of this specification. The detailed performance requirements of the multiple welding sets shall be as specified (see 6.2.1).

3.8 Marking on plates, charts. All words on plates and charts shall be in the English language. Characters shall be engraved, etched, embossed, or stamped in boldface on a contrasting background.

3.8.1 Nameplate. A nameplate shall be attached to each welding set. Unless otherwise specified (see 6.2.1), the nameplate shall contain the information listed below. If the welding set is a special model, the model designation shall include the model of the basic standard welding set and a suffix identified in the manufacturer's permanent records. The captions

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

listed may be shortened or abbreviated, provided the entry for each caption is clear as to its identity.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Contract Number or Order Number
National Stock Number or Plant Equipment Code
Date of manufacture

3.9 Workmanship. Workmanship of the welding sets and accessories shall be of a quality equal to that of the manufacturer's commercial equipment of the type specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

4.3 Inspection conditions. Unless otherwise herein, all inspections, tests, and examinations shall be performed in the manufacturer's designated indoor test area under the ambient temperature, the relative humidity, and

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:39Z
Check the source to verify that this is the current version before use.

the air pressure existing inside the building at the time the inspections, tests, and examinations are performed.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.6 and all tests in 4.7. Failure of the item to pass the first article examination and all tests shall be cause for rejection.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each item prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of the examination in 4.6, the test in 4.7.1, and the inspection in 4.8. Failure of the item to pass the examination, the test, or inspection shall be cause for rejection.

4.6 Examination. The welding sets shall be visually examined to determine compliance with all requirements of this specification.

4.7 Tests. Except as otherwise specified (see 6.2.1), all tests shall follow the procedures of NEMA EW-1, when applicable.

4.7.1 Operational test. All electrical and mechanical components of the welding sets shall be operated to determine if they will function according to requirements as specified herein.

4.7.2 Welding tests. The assembled components of the welding set shall be properly connected and welding tests performed using the smallest and largest size wires the set is designed for. The performance tests shall demonstrate the capability of the set to start and to operate at continuous duty utilizing the ranges of currents, wire speeds, carriage speeds, and utilizing the feed, spreads, and control of the flux. There shall be no evidence of difficulty in striking or maintaining the arc, in the control of the coordinated actions of the moving components, or in the smooth, continuous, and even flow and distribution of the flux. The welds shall be visually examined for even contour and penetration.

4.7.3 Special tests. When specified (see 6.2.1), the supplier shall perform special tests to indicate the equipment complies with unusual operating conditions or optional equipment requirements as required by the procuring activity.

4.8 Inspection of packaging. Packaging shall be inspected to determine conformance of the requirements of section 5.

5. PACKAGING

5.1 Preservation, packing, and marking. Unless otherwise specified, preservation, packing, and marking shall be in accordance with ASTM-D-3951.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

When specified (see 6.2.1), level A or level B preservation, level A or level B packing, and marking shall be accomplished in accordance with MIL-W-45562.

6. NOTES

6.1 Intended use. The welding sets covered by this specification, when connected to the proper power source, are to be used in performing horizontal submerged arc welding of metals using the consumable wire electrode and protective layer of welding flux method.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Types, sizes, and classes, required (see 1.2).

c. First article, if required (see 3.1).

d. If the welding set is required to be configured in a specific measurement system (US or SI), state required system (see 3.2.3).

e. If measuring and indicating devices are required to be graduated in a specific measurement system (US or SI), state required system (see 3.2.3).

f. Additional safety and health requirements, if required (see 3.2.5).

g. Painting, if different (see 3.3.5).

h. Electrical system and input power, if different (see 3.5.1).

i. Limiting devices, if required (see 3.3.8).

j. Specify and fully describe optional equipment, if required (see 3.6).

k. Unfused flux recovery system or fused flux reclaiming system, if required (see 3.6.1).

l. Multiple welding sets (see 3.7).

m. Nameplate, if different (see 3.8.1).

n. First article inspection, if different (see 4.4).

o. Quality conformance inspection, if different (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:39Z
Check the source to verify that this is the current version before use.

p. Power unit, specify military specification (see 3.5.1).

q. Contact devices, specify (see 3.5.3.2).

r. Beam carriage, specify length (see 3.5.5.2).

s. Specify lengths of cables and hoses for special conditions (see 3.5.7).

t. Test, if different (see 4.7).

u. Special test, specify (see 4.7.3).

v. If preservation, packing, and marking in accordance with (MIL-W-45562) are required, specify level required and specification number (see 5.1).

6.2.2 Contract data requirements. Required technical data (operator's manuals, parts lists, wiring diagrams, foundation and anchor bolt plans, and acceptance test reports) should be specified on a DD Form 1423, Contract Data Requirements List, incorporated into the contract.

6.3 First article. When first article inspection is required, the item to be tested should be the first item offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.4 Safety and health requirements. Paragraph 3.2.5 requires compliance only with those OSHA requirements that concern the welding set itself. It does not require compliance with those OSHA requirements that concern "the welding set in its operating environment" such as noise levels, radiation levels, electromagnetic emissions, noxious vapors, air contaminants, and heat. Since OSHA limits the total hazard level of these hazards in the environment (and does not limit the hazard level of individual welding sets in the environment) the procuring activity is advised to analyze the existing hazard levels in the proposed operating environment, and specify additional requirements that will integrate the new welding sets into its future operating environment. The welding set shall be equipped with all point-of-operation guarding normally furnished as standard on the manufacturer's commercial welding set supplied to the commercial market. Any other additional safety and health requirements should be specified in detail under 6.2.1(f).

6.5 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T04:39Z
Check the source to verify that this is the current version before use.

Custodians:
Army - AL
Navy - SH
Air Force - 99

Review activities:
Army - AR
Navy - CG
DLA - GS
Air Force - 84

User activities:
Army - AR
Navy - AS, CG, MC
DLA - GS

Preparing activities:
DLA - IP

(Project 3431-0172)

Source: https://assist.dla.mil -- Downloaded: 202610-23T04:39Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-W-80117B	Welding Sets, Submerged Arc

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE

Source: https://assist.dla.mil – Downloaded: 2022-10-20T04:00Z
Check the source to verify that this is the current version before use.